

Sadegh Mehrabi

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Senior Embedded Engineer — Senior Hardware Engineer — Senior C++ Engineer

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Experienced Senior Embedded Software Engineer with 8+ years of experience in **Embedded Systems, Firmware Development, Embedded C/C++, RTOS, Bare-Metal Programming**, and **real-time** software solutions. Proven expertise in designing and developing high-performance embedded systems, hardware architectures, and scalable real-time applications across **automotive, aerospace, industrial, and medical** industries. Skilled in hardware-software integration, communication protocols, sensor fusion, power electronics, and real-time data processing, with strong adherence to **MISRA C / C++** guidelines for safe and reliable embedded software development. Adept at working across both hardware and software domains, including embedded firmware, PCB design, system architecture, desktop applications, and validation tools for complex engineering systems.

Skills

- **Embedded & Real-Time Systems**

- Embedded C/C++ development for ARM Cortex-M/R, DSP, and AVR microcontrollers.
- RTOS and bare-metal software development for real-time applications.
- Device driver and HAL development for peripherals and communication interfaces.
- Experience with CAN, LIN, SPI, I2C, UART, Ethernet TCP/IP, Wi-Fi, Bluetooth, and ARINC 429.
- Experience with **Bootloaders, DMA, interrupts, timers, queues, and memory management**.

- **Software Development**

- C, C++, C#, HTML, CSS, JavaScript and object-oriented programming.
- Experience with .NET desktop applications, Qt/Qt Quick, and Unity-based tools.
- Experience with software architecture patterns including MVVM, MVC, and SOLID principles.
- Git, GitLab, GitHub, CI/CD pipelines, and automated build systems.

- **Power Electronics**

- Power electronics fundamentals and converter topologies (Buck, Boost, Flyback, Full-Bridge).
- Analysis, simulation, and hardware design of automotive-grade AC-DC and DC-DC converters for EV applications.
- Design of 3.3kW, 400V isolated AC-DC converter for electric vehicles.
- Design of 2.4kW, 12V isolated DC-DC converter for electric vehicles.
- Protection systems design including OVP, UVP, OCP, OTP, inrush current limiting (ICL), short-circuit protection, and EMI filters.
- Simulation and validation using MATLAB/Simulink, PSIM, PSpice, and hardware testing.

- **Electronic Hardware Design**

- Analog and digital circuit design and simulation.
- PCB schematic and multilayer layout design using Altium Designer.
- EMI/EMC considerations, impedance control, and signal integrity techniques.

- **Control, Signal Processing & Robotics (Background)**

- Design and implementation of control algorithms including PID, LQR, and Fuzzy controllers.
- Motor control for DC and stepper motors (position, speed, and torque control).
- Sensor fusion and filtering algorithms including Kalman Filter, EKF, UKF, and complementary filters.
- Simulation and analysis using MATLAB/Simulink and real-time testing environments.

- **Engineering Tools**

- MATLAB/Simulink, PSIM, PSpice, Proteus, Altium Designer, LabVIEW.
- Visual Studio, Qt Creator, STM32CubeIDE, Code Composer Studio, Keil, IAR.

Experience

May.2023–Present
Full time

Software developer

Luminocity AG , Switzerland / Remote

- Developed real-time data acquisition and processing systems for multi-sensor devices (GNSS, IMU, magnetometer, barometer, and camera triggers), handling synchronized sensor streams using **FreeRTOS** with more than 30 concurrent tasks.
- Built cross-platform desktop and web applications using **.NET MAUI** and **Blazor** for real-time monitoring, visualization, and analysis of hardware-generated sensor data.
- Designed modular embedded software architectures for real-time systems with emphasis on performance, scalability, and maintainability.
- Developed **C#** and **C++** components for system integration and data processing.
- Contributed to **CI/CD** workflows using **GitLab** and **GitHub** for automated build, testing, and deployment processes.
- Implemented unit and integration testing using **xUnit** and **Nunit** to ensure system reliability.

Sep.2020–Apr. 2023
Full time

Software developer & Hardware designer

PishroEdehAraz Co. Tehran, Iran

- Designed hardware systems including **Schematics** and **PCB layouts** for embedded and power electronics applications, focusing on reliability, EMI mitigation, and industrial-grade operation.
- Developed embedded firmware using **FreeRTOS** for real-time control and communication systems, including **Modbus TCP/RTU** interfaces for ARINC 429 converters and throttle control systems using **ARM Cortex-M7**.
- Designed and analyzed **automotive-grade AC-DC** and **DC-DC** power converters for electric vehicle applications with emphasis on efficiency, thermal performance, regulation, and system reliability.
- Developed isolated power converter, including a 3.3 kW 400 V AC-DC converter and a 2.4 kW 12 V DC-DC converter using **TI-DSP** for EV
- Implemented protection and safety mechanisms including OVP, UVP, OCP, OTP, short-circuit protection, inrush current limiting (ICL), and EMI filtering for high-reliability power systems.
- Performed simulation, validation, and hardware verification using **MATLAB/Simulink**, **PSIM**, and **PSpice** to evaluate converter performance and control strategies.
- Applied power electronics topologies including **Buck**, **Boost**, **Flyback**, and **Full-Bridge** converters in system design and optimization.
- Designed communication interfaces including **RS485** and **Ethernet TCP/IP** for integration between embedded controllers and power electronics subsystems.

Jun.2018– Sep. 2020
Part time

Firmware developer & Hardware designer

AradSharif Co. Tehran, Iran

- Implemented Ethernet communication protocols including **TCP**, **UDP**, **IP**, and **ICMP** on **STM32** microcontrollers using **LwIP** and **RL-TCPnet** stacks.
- Designed and developed secure **bootloader** solutions with firmware decryption capabilities for reliable and secure microcontroller firmware updates.
- Designed schematics and **6-layer** PCB layouts for complex embedded and instrumentation systems.
- Implemented multitasking and multithreading architectures using **FreeRTOS** for real-time embedded applications.

Software developer & Hardware designer

Kharazmi Co.(Redronic) Tehran, Iran

- Led definition, design, delegation, and implementation of embedded hardware solutions for **AHRS** (Attitude and Heading Reference Systems) with accuracy of 0.1° in roll and pitch and 0.5° in yaw.
- Designed and implemented magnetic disturbance detection algorithms for magnetometer sensors and online and offline sensor calibration.
- Implemented sensor data fusion algorithms including **Kalman filter** (KF), EKF, MAH, and MAD.
- Applied GPS information processing for navigation and integration with **INS**.
- Designed INS hardware and contributed to system integration.
- Designed, modeled, and controlled a **6-RSS parallel robot** as a laboratory prototype.
- Used hardware-in-the-loop (**HIL**) and software-in-the-loop (**SIL**) with MATLAB for validation.
- Designed, modeled, and controlled a self-balancing robot and validated **PID** and digital **LQR** controllers.

Education

- **Master of Science (MSc), Electrical Engineering (Control)** Sep. 2015 – Jan. 2018
 - Amirkabir University of Technology, Tehran, Iran.
 - GPA: **17.7/20** (32 credits) · Ranked **4th** in Electrical Engineering
- **Bachelor of Science (BSc), Electrical Engineering (Control)** Sep. 2010 – Sep. 2014
 - Qom University of Technology, Qom, Iran.
 - GPA: **17.95/20** (141 credits) · Ranked **1st** in Electrical Engineering

Notable Projects

❖ **Mobile mapping system (2025)**

Experienced as a Firmware Developer, Desktop Application Developer, and Hardware Team Lead in the development of a high-precision real-time aerial mapping and navigation system based on dual-core **STM32H7** microcontrollers. Developed embedded firmware using **FreeRTOS** with more than 30 concurrent real-time tasks executing under microsecond-level timing constraints. Integrated multiple sensors including **accelerometers**, **gyroscopes**, **magnetometers**, **barometers**, and dual **GNSS** modules to achieve accurate positioning and orientation estimation for aerial mapping applications. Implemented high-speed interrupt-based data acquisition and synchronized sensor logging to **SD card** storage, alongside **external flash** memory management for system configuration and parameters. Designed and integrated communication interfaces including **Bluetooth**, **Wi-Fi**, and **Ethernet** (LAN) for real-time connectivity and data transfer. Developed precise multi-camera synchronization and triggering mechanisms supporting up to 6 simultaneous cameras for aerial **photogrammetry** and 2D mapping generation. In parallel, developed Windows desktop applications using **.NET Blazor**, **C#**, **JavaScript**, **HTML/CSS**, and **SQLite** for real-time monitoring, visualization, processing, and analysis of sensor and navigation data. Also led hardware development activities including system architecture, hardware integration, debugging, and coordination of embedded hardware design tasks, with strong focus on real-time reliability, synchronization accuracy, and high-performance system operation.

❖ **Medical Auto Sampler system aligned with IEC 62304 (2024)**

Worked as Firmware Developer and Hardware Designer in the development of an automated medical autosampler system for laboratory blood sample processing. Designed the embedded hardware and developed firmware in **Embedded C++** based on **STM32F4** microcontrollers using **FreeRTOS** for reliable real-time operation. The system included a custom mechanical platform capable of holding up to 30 blood sample tubes and performing fully automated sample handling workflows.

Integrated a touchscreen user interface and **USB HID** keyboard connectivity, allowing operators to configure sample quantities and control the sampling process efficiently. Implemented **RS232** communication with ProLYTE medical analysis devices, enabling automated command transmission, synchronization with sampling operations, and retrieval of analysis results directly from the laboratory equipment. Developed reliable data management functionality with storage capacity for up to 300 patient records and test results.

The project required robust real-time task management, hardware–software integration, and stable serial communication to ensure accurate and uninterrupted medical laboratory operation. The system was successfully manufactured and deployed in large quantities for laboratory and medical diagnostic environments.

❖ **Arinc429 converter system (2023)**

Worked as Firmware Developer and Hardware Designer in the complete development of a high-speed **ARINC429 to Modbus TCP** protocol converter for aerospace applications in **Embedded C++** using **STM32H7** microcontrollers. Designed the hardware architecture, schematics, and embedded firmware for a real-time communication system capable of interfacing directly with aircraft ARINC429 buses. The system supported 32 independent ARINC429 receive channels and 16 transmit channels, allowing simultaneous communication with multiple aircraft subsystems and engine control units. Implemented high-performance bidirectional data conversion between ARINC429 and Modbus TCP over 100 Mbps Ethernet, enabling real-time data exchange between desktop software and aircraft engines. Developed the firmware using **FreeRTOS** with approximately **25** concurrent real-time tasks, with strong focus on deterministic execution, low-latency communication, and zero data loss operation. Designed the system to handle continuous high-speed data streams reliably under strict real-time constraints, ensuring robust synchronization and stable operation in mission-critical aerospace environments.

❖ **On board converter system for Electric Vehicle (2022)**

Worked as Embedded Software Developer and Hardware Designer in the development of an automotive-grade EV charger supporting **single-phase** (3.3 kW) and **three-phase** (10 kW) operation. Designed full power electronics hardware including schematic design, transformer/inductor calculations, control stage architecture, simulation, and **8-layer PCB** layout with focus on **EMC/EMI/ESD** compliance (**ISO 7637**, **ISO 16750**) and functional safety considerations aligned with **ISO 26262** principles.

Implemented embedded control firmware in **Embedded C++** on **TI C2000 microcontrollers** using bare-metal architecture for deterministic real-time performance. Developed and tuned digital control algorithms including **Adaptive PID** for power regulation, applying advanced topologies such as **Phase-Shifted Full-Bridge (PSFB)** and **Interleaved PFC** for efficiency and thermal optimization. Integrated **CAN 2.0** communication (**ISO 11898**) for real-time vehicle–charger interaction and system monitoring.

Additionally developed a **Qt Widgets** desktop application for testing, debugging, and validation via serial-to-CAN interface. The project required end-to-end integration of embedded control, power electronics, and communication systems for high-reliability automotive EV charging systems.

❖ **Attitude and Heading Reference System (AHRS) (2021)**

Worked as Firmware Developer and Hardware Designer in the development of a high-precision AHRS (Attitude and Heading Reference System) for real-time orientation and motion estimation applications. Designed the embedded hardware and developed firmware using **Embedded C and Bare-Metal programming** for sensor acquisition, signal processing, and real-time attitude estimation using gyroscope, accelerometer, and magnetometer sensor fusion. Implemented advanced estimation and filtering algorithms including **Kalman Filter-based** approaches to accurately calculate Euler angles, quaternions, and angular velocity under dynamic operating conditions.

Developed complex calibration and compensation algorithms for sensor alignment, bias correction, and magnetic calibration to improve long-term stability and measurement accuracy. Designed and integrated desktop software for real-time data visualization, logging, monitoring, and execution of calibration procedures through direct hardware communication. Achieved high-accuracy orientation estimation with **roll** and **pitch** accuracy reaching **0.1°** and **yaw (heading)** accuracy of **0.5°**. The project required strong expertise in embedded systems, mathematical modeling, control systems, real-time signal processing, and hardware–software integration.

** More details and visual examples of my projects are available on my website:*

<https://sadegh-mehrabi.github.io/resume>